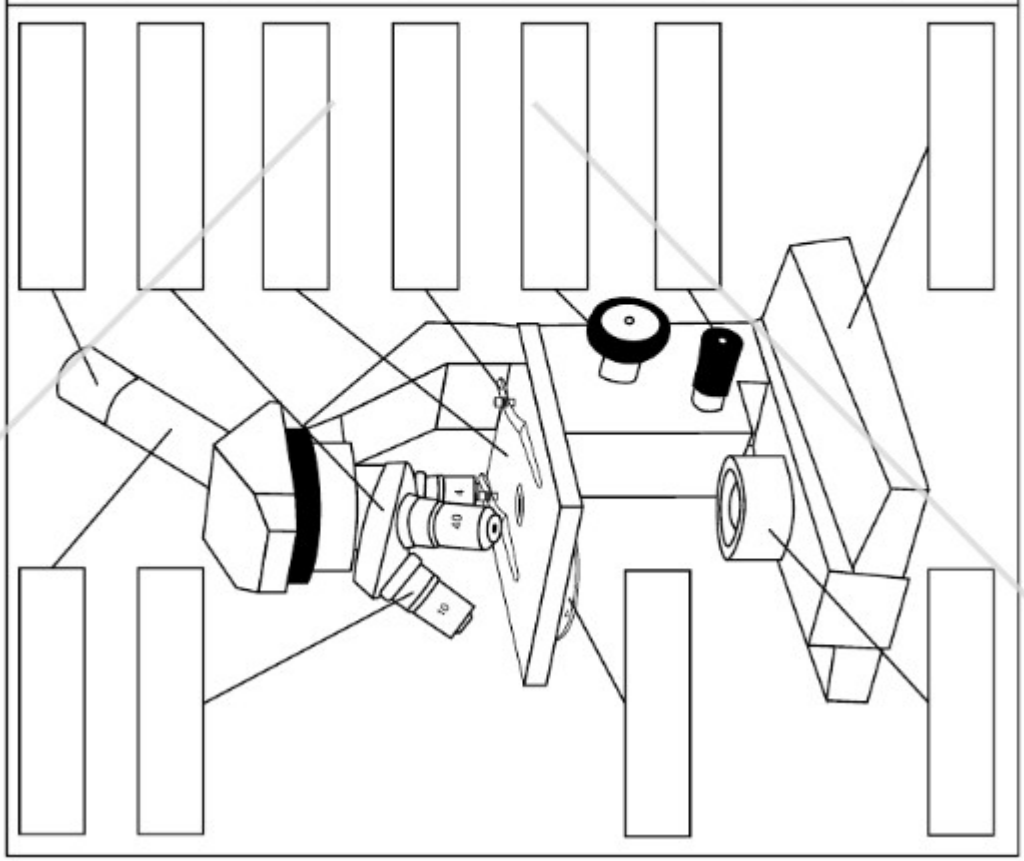
Shaped for the job

Name_____ Class_____ Date_____

Cells look differently because they do different jobs.
Draw lines to mach up the cell, it's job and the link.

Cell	Job	Link
 root hair cell	deliver oxygen around body	'Hair like' structure: increases surface area
 sperm cell	absorb water from the soil	No nucleus, so it has an increased surface area
 Red blood cell	it carries signals around the body	Tail helps it swim, lots of mitochondria for energy
 Neuron	fertilise the ovum (egg) cell	Very long cells so that signals can be carried quickly around the body

Microscope Part	Eye Piece	Objective Lens	Nose Piece	Stage	Stage Clips	Rough Focus Wheel	Fine Focus Wheel	Iris Diaphragm	Light



PARTS OF THE CELL

Label the cell parts in the diagrams below

1. Animal Cell

2. Plant Cell

Cell Structure
Nucleus
Cytoplasm
Cell surface membrane
Mitochondria
Cell wall
Chloroplast
Vacuole

For each level of organisation give an example. One example has been done for you:

Cell ⟶ Tissue ⟶ Organ ⟶ Organ System ⟶ Organism

e.g. Palisade cell	e.g. Palisade mesophyll	e.g. Leaf	e.g. Shoot system	e.g. Plant

Not all organisms show all of the above levels of organisation. Some organisms exist as single cells, these are known as u _ _ _ _ _ _ _ _ _ _ _ e.g. bacteria. Animals and plants however are made from many cells and are known as m _ _ _ _ _ _ _ _ _ _ _ _ organisms.

Plant and Animal Cells

Activity 1: Which bit is it?



Which part ...

1. ...of an animal cell controls the cell's activities? .....
2. ...of an animal forms a thin layer around the outside? .....
3. ...is a liquid found inside animal cells? .....
4. ...is a green structure found inside plant cells? .....
5. ...is a sac of liquid found inside plant cells? .....

Activity 2: True or False?

Read each statement and decide whether it is true or false:

1. Almost all cells have a nucleus. .....
2. All plant cells contain chloroplasts. .....
3. The cell membrane controls what goes in and out of the cell. .....
4. Animal cells have cell membranes. .....
5. Animal cells have cell walls. .....
6. All cells contains some cytoplasm. .....

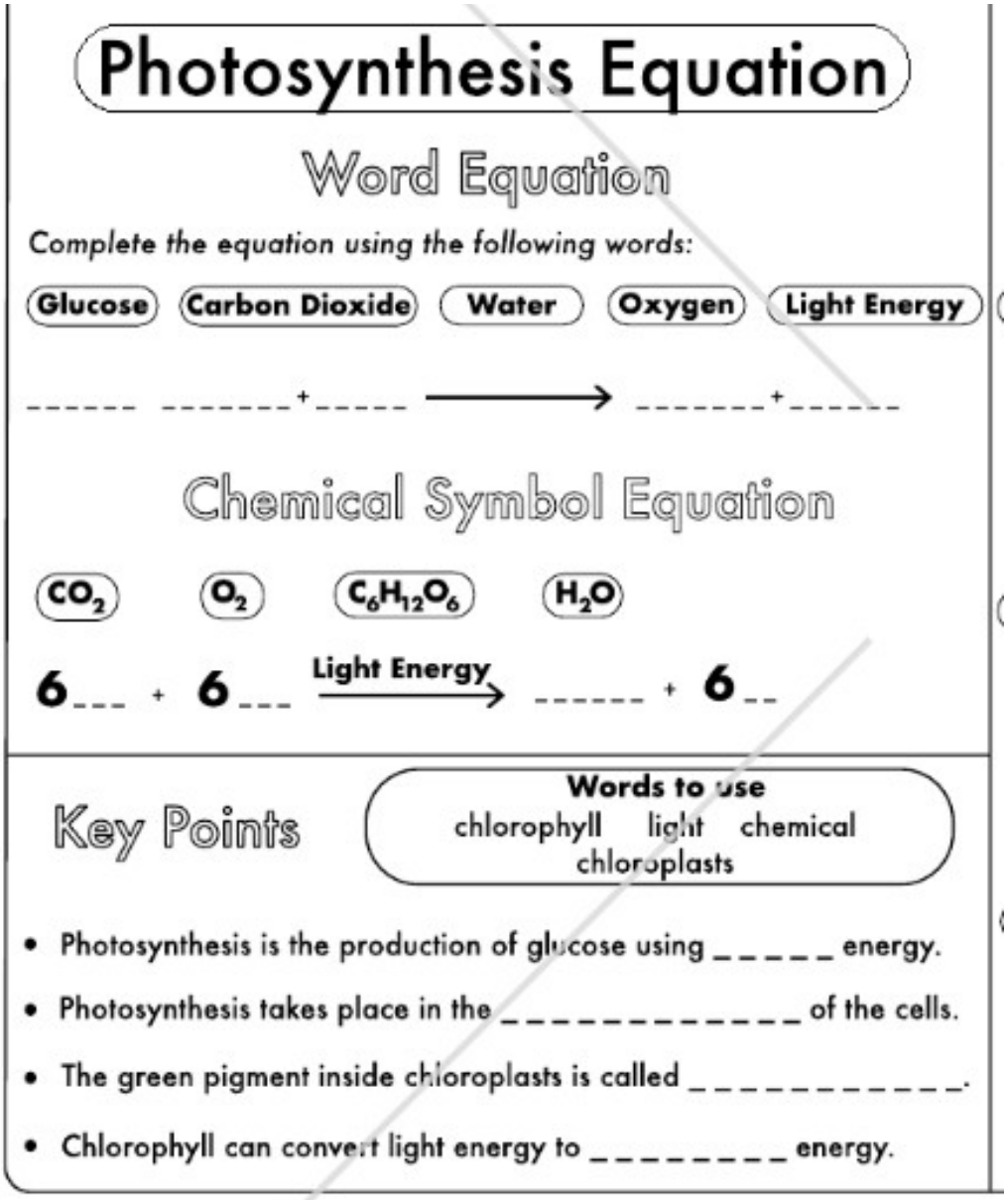
Activity 3: Explanations

Match each statement with the correct explanation.

Statement	because	Explanation
Plant and animal cells have many features in common		it is needed to control what goes in and out of the cell
There are some differences between animal and plant cells		they need to perform different functions
Plant cells have a cell wall		they are unable to photosynthesise
Both plant and animal cells have a cell membrane		they need to perform the same functions
Animal cells do not have chloroplasts		it contains the instructions that control what the cell does
Nearly every type of cell has a nucleus		it gives the cell shape and supports the plant

Draw the following organelles:

Golgi body	
Mitochondria	
Rough endoplasmic reticulum	



Instructions: Use the clues in each question to identify whether or not that scenario is describing photosynthesis or cellular respiration.

#1. This happens in the chloroplast. _____	#2. It requires light energy. _____	#3. It produces carbon dioxide (CO_2). _____
#4. It produces oxygen (O_2) _____	#5. One reactant is oxygen (O_2) _____	#6. One reactant is glucose _____
#7. It does not require light energy. _____	#8. It produces glucose. _____	#9. It produces chemical energy (ATP). _____
#10. It needs carbon dioxide (CO_2) from the air. _____	#11. This happens in the mitochondria. _____	#12. Only plants can perform this reaction. _____
#13. Both plants and animals can perform this reaction. _____	#14. It makes water. _____	#15. It needs water. _____

Pictionary:

Mitochondria

Golgi body

Endoplasmic reticulum

DNA

Respiration

Photosynthesis

Animal cell

Plant cell

Microscope

Tissue

Organ

Organ system

Multicellular

Unicellular